Activity 8 - Quantify and Audit the Product Carbon Footprint Model

Alignment: K2 | K4 | K5 | A3 | A4 | LO3 Duration: 55 minutes Objective: Use a controlled spreadsheet/software workflow to quantify the Lumina 400 footprint and test its model integrity.

Scenario

HarbourLight's current model mixes kg and tonnes, uses an undated China grid factor, links one aluminium dataset with recycled content inconsistent with procurement, and applies a 25-year product lifetime although the declared rating is 50,000 operating hours. A hidden formula omits air freight from the total, while the report rounds the final result to one decimal place.

Materials

- Activity 7 lifecycle inventory
- Illustrative factor and GWP register
- Seeded calculation model
- Software setup and audit checklist

Questions

- Which model settings and datasets must be declared?
- What control failures affect the total?
- Which lifecycle stages dominate and why?
- How should corrections and residual limitations be reported?

Detailed procedure

1. Create a controlled factor register with source, version, geography, technology, unit, GWP basis and approval.
2. Standardise activity units and document every conversion before applying factors.
3. Calculate each flow using activity data, factor and any required gas-to-CO2e conversion.
4. Aggregate results by unit process, lifecycle stage, gas and relevant scope/category.
5. Record software or workbook settings, functional unit, allocation, datasets and model version.
6. Run contribution analysis and identify high-impact model links for detailed review.
7. Trace selected results backward to source records and forward from source to report.
8. Reperform high-risk rows, find seeded unit, factor, lifetime and formula defects, then correct them under change control.
9. Prepare an old-to-new result bridge and disclose residual estimates, uncertainty and limitations.

Required evidence

- Controlled factor register
- Quantified product footprint
- Software/model setup record

- Audit tests and correction bridge

Acceptance check

The model is reproducible, units and factors are controlled, seeded defects are resolved, contribution results reconcile, and the correction bridge preserves a complete audit trail.

Files

- Activity-08-Workbook.xlsx - editable working file
- Activity-08-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

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